

MN 4023 - Individual Assignment

FERROCEMENT WALL PANELS FOR CONSTRUCTION

Group Members

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Introduction

- **Product:**
- Prefabricated ferrocement wall panel made of a cement-sand mortar reinforced with wire mesh, suitable for modular walls in toilets, tanks, and housing units.
- **Existing Issues:**
- Current prototypes at NERDC face cracking during lifting/handling and excessive self-weight, making transport and installation difficult.
- **Final Goal:**
- To create an optimized, durable, and sustainable wall panel - reducing weight, preventing cracks.



TECHNICAL FEASIBILITY

Resources

Materials	River sand, M-sand, fly ash, bottom ash, rice husk ash	
Reinforcement & connections	wire mesh, mild steel plates, tor steel	
Equipment	mixing, moulding, curing facilities, masonry tools	



1. Reinforcement layout



2.Cured panels

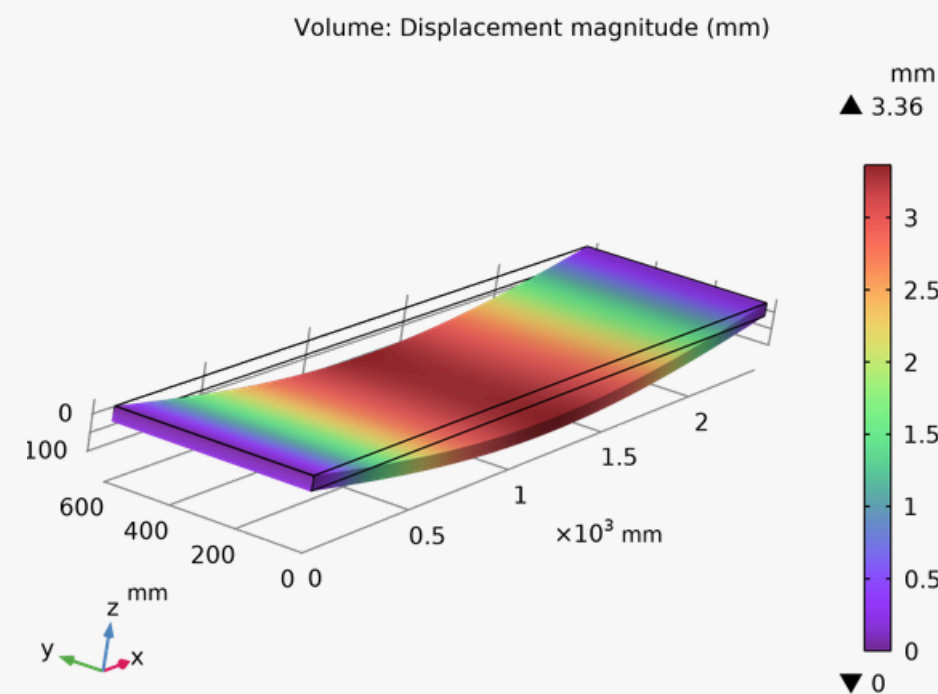
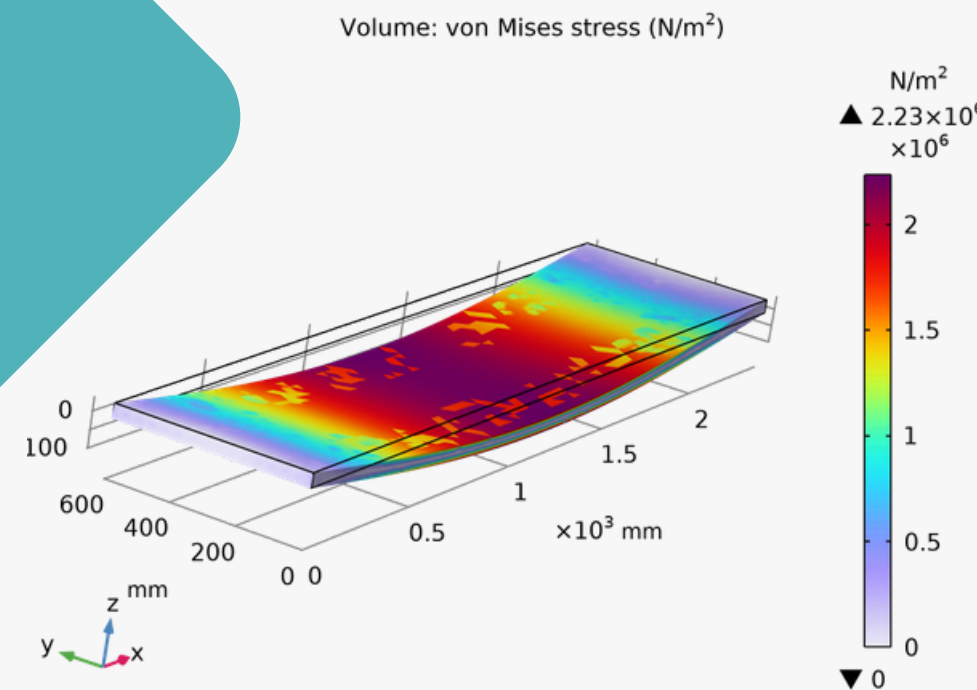
Method

- COMSOL linear-elastic static analysis
- simply supported at short edges
- load = self-weight.

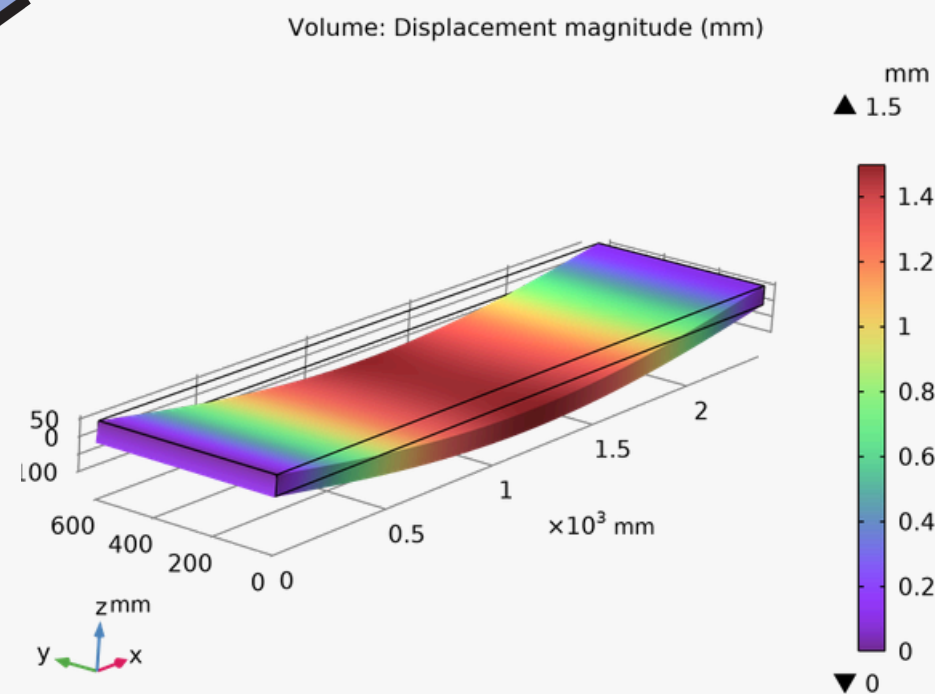
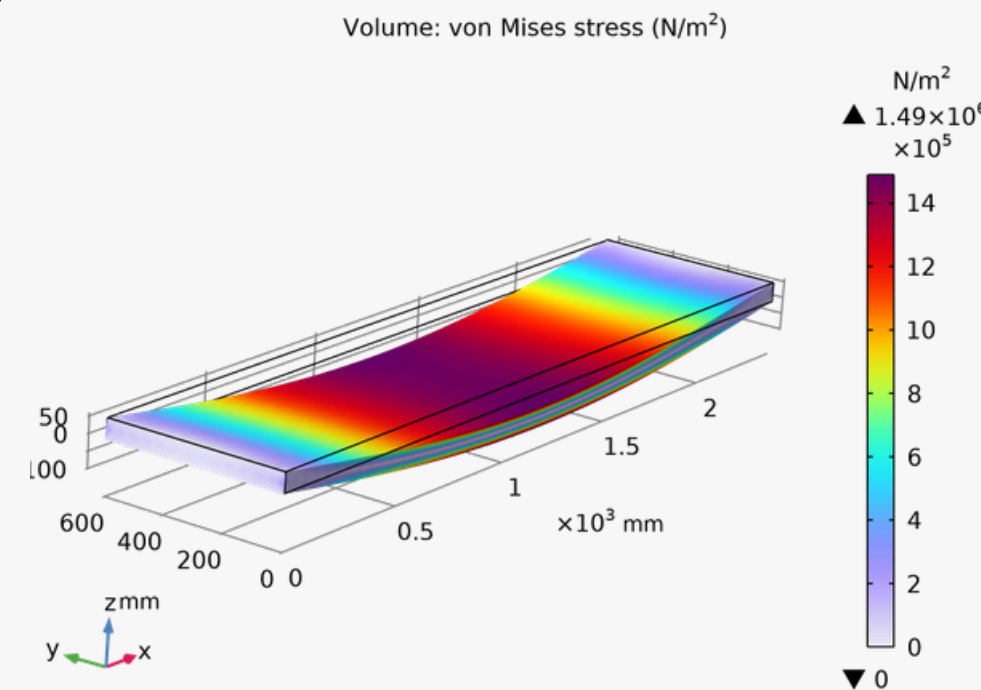
Expertise

- Institutes: NERDC, UoM
- Human capability
- Technology transfer

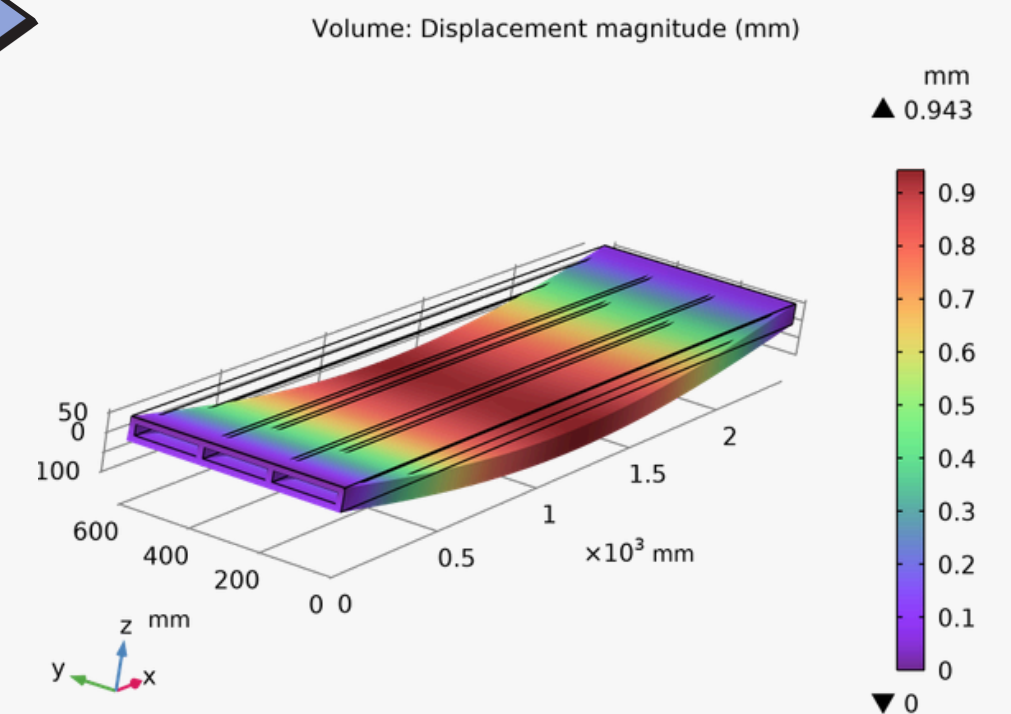
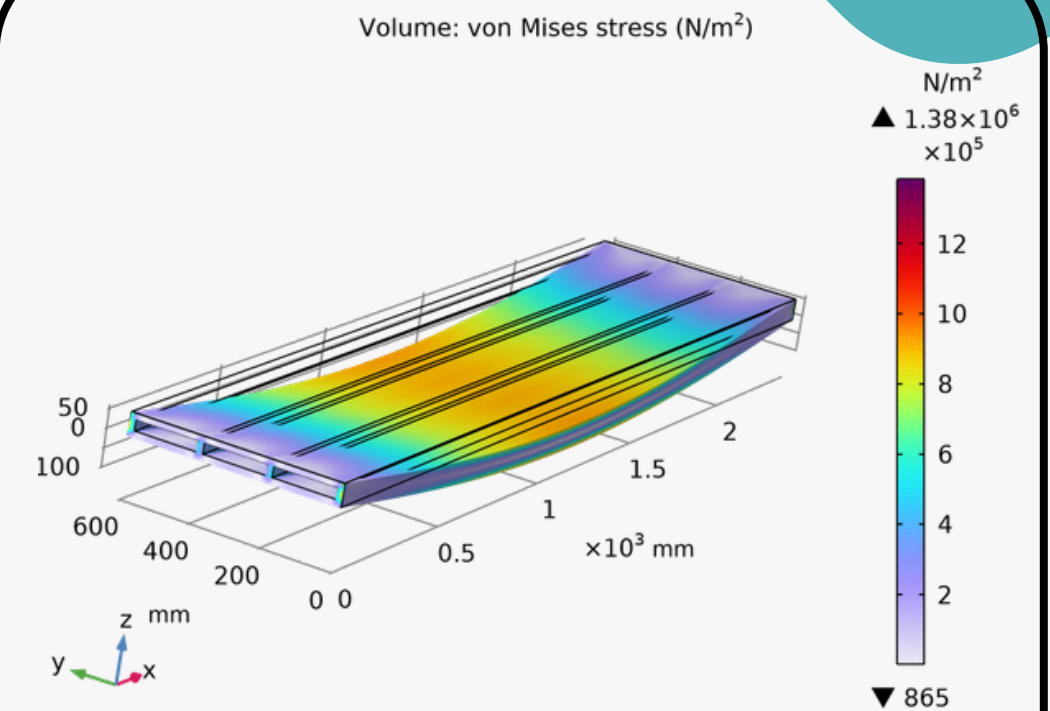
FE Analysis



a) 40 mm panel (solid)



b) 60 mm panel (solid)



c) 60 mm 3-core panel



Financial Analysis

Cash Outflows

- Capital Expenditure
- Operational and Maintenance Costs
- Tax
- Total Cost per panel ~ 4200 LKR

Cash Inflows

- Panel Sales
- Salvage Value
- Selling Price of a panel = 7000 LKR

C) Equipment & CapEx (one-time)	
Mold price (LKR)	5,000
Molds subtotal (LKR)	630,000
1-bag site mixer (≈280–420 L) (LKR)	350,000
Poker/needle vibrator 35 mm (LKR)	19,500
Other masonry tools (LKR)	50,000
Curing/storage racks (LKR)	600,000
Yard (LKR)	2,000,000
Curing hut (LKR)	150,000
Total CapEx (LKR) yr 0	3,804,500
D) Per-Panel Variable Costs & Logistics	
Material cost (LKR/panel)	2,500
Water supply (LKR/panel)	40
Electricity (LKR/panel)	40
Install/transport (LKR/panel)	200
Variable cost inflation per yr	5%
Total variable cost (LKR/panel)	2,780

Year	Total	0	0.5	1	1.5	2	2.5	3	3.5	4
Financial Inflows										
Panel sales revenue (LKR)	94,650,526.09	0.00	0.00	3,911,040.00	4,399,920.00	4,531,917.60	4,531,917.60	4,667,875.13	4,667,875.13	4,807,911.38
Salvage value (LKR)	2,180,450.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Financial Inflows (LKR)	96830976.09	0.00	0.00	3911040.00	4399920.00	4531917.60	4531917.60	4667875.13	4667875.13	4807911.38
Financial Outflows										
Capital expenditure (CapEx) (LKR)	3,804,500.00	3,804,500.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Production (materials + direct labour + QA + utilities + install/transport)	50,119,355.93	0.00	2,213,252.77	2,455,557.56	2,455,557.56	2,504,668.71	2,504,668.71	2,554,762.08	2,554,762.08	2,605,857.33
Fixed O&M (maintenance, supervision, tools)	2,000,359.78	0.00	91,225.00	93,961.75	93,961.75	96,780.60	96,780.60	99,684.02	99,684.02	102,674.54
Total Financial Outflows (LKR)	55924215.71	3804500	2304477.77	2549519.31	2549519.31	2601449.31	2601449.31	2654446.1	2654446.1	2708531.87
Net Financial Cash Flow (LKR)	40906760.39	-3804500	-2304477.77	1361520.69	1850400.69	1930468.29	1930468.29	2013429.02	2013429.02	2099379.51
Tax (LKR)	11283777.16	0	0	326764.966	444096.166	463312.389	463312.389	483222.966	483222.966	503851.084
Net Financial Cash Flow after tax (LKR)	29622983.23	-3804500	-2304477.77	1034755.73	1406304.53	1467155.9	1467155.9	1530206.06	1530206.06	1595528.43
3.Present Value Calculations										
PV of Inflows (LKR)	52499884.92	0	0	3492000	3712083.58	3612816.96	3413791.15	3322501.32	3139468.65	3055514.6
PV of Outflows (LKR)	34233495.73	3804500	2177526.82	2276356.53	2150954.74	2073859.46	1959613	1889382.31	1785298.47	1721320.97
PV of Net inflows (LKR)	12446769.35	-3804500.00	-2177526.82	923889.04	1186457.92	1169607.70	1105175.39	1089170.45	1029169.34	1013987.16
Cumulative PV Net (LKR)		-3804500.00	-5982026.82	-5058137.78	-3871679.86	-2702072.16	-1596896.76	-507726.32	521443.02	1535430.18

Performance Indicators

- NPV: 12.45 LKR mn
- IRR: 26.36%
- BCR: 1.204
- Disc. PBP: 3.25

4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
4,807,911.38	4,952,148.72	4,952,148.72	5,100,713.18	5,100,713.18	5,253,734.58	5,253,734.58	5,411,346.62	5,411,346.62	5,573,687.02	5,573,687.02	5,740,897.63
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2,180,450.00	0.00
4807911.38	4952148.72	4952148.72	5100713.18	5100713.18	5253734.58	5253734.58	5411346.62	5411346.62	5573687.02	7754137.02	5740897.63
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2,605,857.33	2,657,974.47	2,657,974.47	2,711,133.96	2,711,133.96	2,765,356.64	2,765,356.64	2,820,663.77	2,820,663.77	2,877,077.05	2,877,077.05	0.00
102,674.54	105,754.78	105,754.78	108,927.42	108,927.42	112,195.24	112,195.24	115,561.10	115,561.10	119,027.93	119,027.93	0.00
2708531.87	2763729.25	2763729.25	2820061.38	2820061.38	2877551.88	2877551.88	2936224.87	2936224.87	2996104.98	2996104.98	0
2099379.51	2188419.47	2188419.47	2280651.8	2280651.8	2376182.7	2376182.7	2475121.74	2475121.74	2577582.03	4758032.03	5740897.63
503851.084	525220.674	525220.674	547356.433	547356.433	570283.847	570283.847	594029.218	594029.218	618619.688	1141927.69	1377815.43
1595528.43	1663198.8	1663198.8	1733295.37	1733295.37	1805898.85	1805898.85	1881092.53	1881092.53	1958962.35	3616104.35	4363082.2
2887189.92	2809982.18	2655183.58	2584180.04	2441820.62	2376522.72	2245602.89	2185552.14	2065152.66	2009927.41	2642179.12	1848415.39
1626495.43	1568214.2	1481823.13	1428730.86	1350023.77	1301658.34	1229951.52	1185891.98	1120562.59	1080425.49	1020906.13	0
958127.81	943743.67	891753.94	878141.38	829765.61	816896.93	771895.04	759741.72	717888.45	706421.46	1232167.47	1404795.70
2493557.99	3437301.66	4329055.60	5207196.98	6036962.58	6853859.51	7625754.55	8385496.27	9103384.72	9809806.18	11041973.65	12446769.35



Economical Analysis

Costs

- Capital Expenditure
- Materials & Labour
- Operation & Maintenance
- Environmental Management

Benefits

- Panel Sales
- Employment
- Environmental Benefits
- Time Saving
- Salvage Value

Performance Indicators

- NPV: 30.1 LKR mn
- IRR: 37.4%
- BCR: 1.8792
- Disc. PBP: 1.83

Multi Criteria Analysis



Criterion	Clay Brick	Concrete Brick	RC Wall Panel	Ferrocement Panel
Cost (40%)	3.5	4	3	5
Time/speed (25%)	1.5	2	4	5
Environmental (15%)	2	2.5	2	3.5
Durability (20%)	5	4	3	3.5
Weighted Score	3.075	3.275	3.100	4.475

Sensitivity Analysis



Main Factors

- Selling Price
- Material Cost
- Discount Rate

Parameter	Base Value	Range Tested (Min)	Range Tested (Max)	Base FNPV (LKR Mn)
Material cost (LKR/panel)	2,500	2000	3000	12.45
Selling price (LKR/panel) yr 1	7000	5600	8400	12.45
Discount rate (EAR)	12%	8%	16%	12.45
O&M % yr ≥ 0.5 from tot.CapEx	10%	8%	12%	12.45
steady-state utilization (Year ≥ 0.5)	90%	80%	100%	12.45
Product & line scrap/reject rate	3%	1%	5%	12.45
Project duration (years)	9.5	7.5	11.5	12.45
Total CapEx (LKR) yr 0	3799500	3039600	4559400	12.45
Variable cost inflation per yr	2%	1%	5%	12.45

FNPV @ Low (LKR Mn)	FNPV @ High (LKR Mn)	FNPV Frac Change	Sensitivity Index (E)	Sensitivity Level
15.41	9.49	0.48	0.1902	Low
4.59	20.32	1.26	0.5054	Moderate
16.52	9.40	0.57	0.3813	Low
12.63	12.27	0.03	0.0116	Low
9.62	14.34	0.38	0.0842	Low
12.93	11.98	0.08	0.1017	Low
9.60	14.05	0.36	0.1505	Low
13.10	10.87	0.18	0.0716	Low
12.60	9.95	0.21	0.4257	Low

Environmental Analysis



Benefits

- Lower resource consumption
- Reduced embodied energy
- Less on-site waste
- Reusing waste
- Less noisy

Drawbacks

- Cement footprint
- Mesh corrosion waste
- Transport emissions
- End-of-life recycling challenge
- Water use in curing

Social Analysis



Benefits

- Employment diversification
- Reskilling opportunities
- Affordable housing
- Community uplift
- Cleaner and safer sites
- Time efficiency

Drawbacks

- Labor displacement
- Cultural resistance
- Skill gap
- Quality inconsistency risk
- Urban–rural imbalance
- End-of-life waste

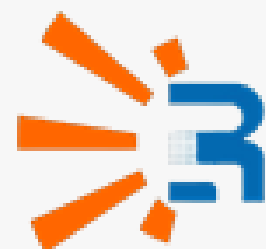
Risk Assessment



Category	Main Risks	Severity	Mitigation Strategy
Technical	Panel cracks, mesh corrosion	High	Design optimization, protective coatings, handling trials
Testing / Process	High testing cost, durability uncertainty	Medium	Lab partnerships, accelerated testing
Market & Social	Low demand, public reluctance	High	Awareness campaigns, pilot houses
Regulatory	Approval delays	Medium	Early engagement, certifications
Health & Safety	Worker injury	High	PPE, lifting guides, safety training
Environmental	Waste generation	Low	Reuse and recycling
Aesthetic	Surface finish defects	Low	QC and surface coating

Final Recommendation

- The ferrocement wall panel development project is technically sound, financially viable, and economically beneficial.
- It reduces cost, time, and CO₂ emissions while creating local employment.
- Recommended for implementation and scale-up under NERDC–UoM collaboration.



**National Engineering Research &
Development Centre of Sri Lanka**





THANK
YOU

